An interpretable Dictionary Learning approach for Mult-class abnormality segmentation from 3D Chest CT

Abstract

Automated Diagnostic Hypotheses from 3D Computed Tomography (CT) requires clinically interpretable methods to support radiological decision making. Current state-of-the-art (SotA) deep learning models offer high classification accuracy, but often rely on post-hoc explainability techniques that provide approximate localizations and lack direct traceability to the models decisions. Multi-class segmentation models exist, but SotA approaches rarely support simultaneous multi class classification and grounded localization, leading to a fragmented approach to diagnosis. This also hinders the ability to benchmark models across multiple dimensions of performance, including classification accuracy, segmentation quality, and explainability. This study aims to address these challenges through two contributions: First, we introduce an inherently interpretable dictionary learning framework in which detected abnormalities can be back-projected to the original voxel space, allowing diagnostic decisions to be traced back to relevant anatomical structures. Second, we develop a unified evaluation framework to benchmark SotA models across three axes: (i) classification accuracy, (ii) segmentation quality, and (iii) explainability, using standardized metrics and a publically available 3D chest CT dataset with expertly annotated abnormality localizations. Our method is benchmarked against SotA models using this framework, demonstrating competitive classification and segmentation performance while providing improved interpretability compared to deep learning baselines.

Chapter 1

Introduction

Developing a unified pipeline for multi-class abnormality classification and grounded localization addresses two of the most critical imperatives in clinical AI adoption: diagnostic accuracy and explainability. Specifically, utilizing dense pixel-level segmentation as the mechanism for localization forces the architecture to provide voxel-level visual evidence for its diagnostic claims. This ensures that the model is held accountable and its reasoning aligns with clinical diagnosis.

The transition from 2D projectional radiography to volumetric CT represents a paradigm shift in medical image analysis. For Lung disease diagnosis, Chest X-Rays (CXR) are commonly used as the first imaging test due to its low cost and low radiation exposure. However, studies show that their capability to detect abnormailities is limited when compared to 3D CT. [1].

Furthermore, CXR has historically benefited from the availability of massive, radiologist-annotated datasets, while the analysis of chest CT volumes has been constrained by a lack of such data. However, the recent introduction of large scale, text-paired datasets has facilitated the development of state-of-the-art (SotA) vision language foundation models tailored specifically for volumetric imaging [2**?** , 3].

Beyond deep learning approaches, Dictionary Learning algorithms, particularly, Label Consistent KSVD (LCKSVD) [4], Label Embedded Dictionary Learning (LEDL) and Fischer Discriminant Dictionary Learning (FDDL) offer viable approaches for interpretable CT analysis. All three techniques incorporate label information during the dictionary training process, and ensures that the sparse representation of the data is highly discriminative, making them well-suited for multi-abnormality tasks.

The clinical utility of Deep Learning based approaches is hindered by a lack of interpretability. While tools like Grad-CAM provide post-hoc explanations to model decisions, these are often insufficient for medical professionals who require a granular understanding of the model's decision making process. In high stakes environments, a model that produces a diagnosis without transparency in its reasoning is difficult to integrate into clinical workflows, as it lacks the accountability necessary for legal and ethical compliance.

1.1 Literature Review

The landscape of 3D CT analysis is currently divided into three primary architectural paradigms:

- Classificationbased models that prioritize diagnostic labels
- Segmentationbased models that prioritize boundary precision
- Unified architectures designed for grounded localization

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SotA models are predominantly specialized in either classification or segmentation. However, recent systems, particularly in 2025 and 2026, are moving toward multitask learning or foundation models that can perform both diagnosis and voxellevel segmentation simultaneously.

Table 1 summarizes the key models in each category, their primary task orientation, and their performance on standard benchmarks.

Table 1.1: Summary of major 3D CT analysis paradigms and representative models

Paradigm	**Representative Models**	**Primary Task**	**Typical Strengths / Notes**
Classificationbased	CTCLIP; Merlin; DINOv3	Imagelevel diagnosis; retrieval; linear probing	Strong imagelevel diagnosis; good zeroshot and transfer performance; efficient inference; limited voxellevel localization.
Segmentationbased	nnUNet; SegVol; SAMMed3D	Voxellevel lesion segmentation	High boundary precision and dense localization; foundation models aim for universal volumetric segmentation across organs and pathologies.
Unified architectures	BiomedParse; MedImageParse; CitrusV; MTMed3D; MedVista3D	Joint diagnosis, segmentation, and reasoning	Multitask inference with grounded localization; integrates report understanding with voxel outputs; typically more complex and computationally intensive.

1.1.1 Classification Architectures

Classificationbased models for 3D CT have shifted away from 3D Convolutional Neural Networks (CNNs) such as 3D UNet, 3D ResNet and DenseNet pipelines, toward Vision Transformers (ViTs) and selfsupervised encoders. The benchmark for this paradigm is the CTRATE dataset, which introduced the first public largescale 3D medical imaging dataset with paired textual reports, enabling the development of Contrastive LanguageImage Pretraining for Computed Tomography (CTCLIP).

CTCLIP is a VLFM designed for 3D chest CT analysis. It extends the CLIP framework to volumetric medical imaging by jointly learning representations from CT scans and their corresponding radiology reports. [5, 6] Through contrastive learning, the model aligns image features with textual descriptions, enabling tasks such as zeroshot classification, crossmodal retrieval, and reportguided abnormality localization. [5] The model is particularly effective in learning semantic relationships between radiological findings and volumetric imaging patterns such as GGO, consolidation, fibrosis, and pulmonary nodules. [5, 7]

Recent benchmarks such as Merlin and DINOv3 have further challenged the necessity of domainspecific pretraining. Merlin is a VLFM exclusively pretrained on abdominal CT scans. Despite it's training, Merlin has consistently demonstrated exceptional zeroshot performance on chest CT classification tasks, even when it was restricted to a linear probe, significantly outperforming models like CTCLIP, M3FM, and CTFM that were explicitly trained on thoracic data, by an average of 12.3% in Area Under the Receiver Operating Characteristic Curve (AUROC). Specifically, in crossdomain retrieval tasks, Merlin surpassed CTCLIP by 24.7%.

DINOv3, a selfsupervised ViT pretrained on over 1.7 billion natural images, outperform specialized medical models like CTNet and CTCLIP across all metrics on several classification tasks due to it's ability to capture finegrained textural features.

1.1.2 Segmentation Architectures

The prevailing SotA for supervised volumetric segmentation remains variations of the 3D UNet, such as nnUNet, which utilizes an adaptive framework to optimize preprocessing and architecture for specific datasets. However, the field has recently shifted toward foundation models capable of universal segmentation such as SegVol and SAMMed3D. The SegVol model is the first 3D foundation model specifically designed for universal volumetric medical image segmentation. Similarly, SAMMed3D adapts the Segment Anything Model (SAM) for 3D medical data.

Recent evaluations published at the Machine Learning for Healthcare (MLHC) Conference in 2025, which specifically utilized the ReXGroundingCT dataset, revealed that the current SotA textprompted segmentation models that demonstrate nearperfect performance in general computer vision tasks struggle with the nuanced clinical descriptions and illdefined boundaries of chest CT abnormalities.

1.1.3 Unified Architectures

Recent models have been designed to integrate abnormality detection, segmentation, and multimodal chainofthought reasoning, enabling pixellevel localization, structured report generation, and physicianlike diagnostic inference in a single framework. BiomedParse, and it's clouddeployed version MedImageParse, developed by Microsoft, are foundation models designed to parse 3D medical images. Other notable models include CitrusV, MTMed3D and MedVista3D, where each architecture incorporates a multitask learning (MTL) framework [8–10], as summarized in Table 1.1.

1.1.4 Dictionary Learning Approaches

Applications of sparse dictionary learning in medical imaging have demonstrated significant success across diverse diagnostic tasks. In histopathological analysis, Discriminative Feature oriented Dictionary Learning (DFDL) has shown the ability to learn class specific dictionaries that achieve high classification accuracy even with limited training data, a critical advantage when generous training samples are not available [11]. In neuroimaging, multi feature kernel supervised dictionary learning approaches have demonstrated superior performance in distinguishing Alzheimerś disease patients from cognitively unimpaired individuals, achieving 98.18% classification accuracy while maintaining fast testing times [12].

Most relevant to our work, dictionary learning has been successfully applied to lung disease diagnosis from CT images. One approach utilized both frozen dictionary learning and label consistent LCKSVD to classify COVID 19 CT scans, achieving 89% accuracy on the CCCCII dataset [13]. The frameworkś key advantage lies in its interpretability, where the sparse coding output can be directly mapped back to the original input signal domain. This work demonstrates that sparse representation methods offer an effective balance between performance and explainability for lung disease classification tasks, providing clear feature representations while maintaining competitive accuracy. Similarly, sparse representation methods have achieved 95.4% accuracy in classifying diffuse lung disease patterns on HRCT images, effectively handling the high variety and complexity of patterns [14].

1.1.5 Explainability Approaches

GradCAM is one of the most widely used posthoc explanation methods for CNNbased classifiers [15]. It generates classdiscriminative heatmaps by using gradients flowing into the final convolutional layer, allowing the user to identify image regions that influenced a prediction [15]. It's main limitation is that it produces lowresolution saliency maps and does not directly provide precise lesion boundaries or voxellevel localization, which is a major drawback for 3D CT interpretation [15, 16].

SegGradCAM extends this idea by adapting GradCAM to semantic segmentation outputs [17]. Instead of only highlighting regions that are important for imagelevel classification, it produces pixellevel relevance heatmaps for segmentation predictions [17]. This is particularly useful for volumetric medical imaging tasks where the target abnormality occupies a sparse region within a large 3D scan.

HiResCAM addresses one of the key weaknesses of GradCAM by producing higherresolution and more faithful localization maps [16]. This is important in chest CT because many abnormalities, such as small nodules, consolidations, or subtle interstitial changes, may be missed by coarse heatmaps. HiResCAM aims to preserve fine spatial detail, making it more suitable for medical imaging applications where precise localization is required [16]. However, like other gradientbased methods, it remains a posthoc explanation technique and does not change the internal decision process of the model.

Chapter 2

Methodology

We implement a sixstage pipeline for our study, consisting of:

1. Data Acquisition
2. Data Preprocessing
3. Implementation of the LCKSVD Approach
4. Abnormality Classification
5. BackProjecting ClassDiscriminative Atoms and Deriving Explainability through Segmentation Masks
6. Evaluation and Benchmarking

2.1 Data Acquisition

Two public chest CT datasets: CTRATE[18] and ReXGroundingCT[7] were used in our work. CTRATE was used as the primary source of CT volumes (normal and abnormal). while ReXGroundingCT was used to obtain voxellevel segmentation masks per abnormality scan together abnormality labels, for each corresponding CT volume. Although CTRATE also contains abnormality labels, RexGroundingCT was chosen for its more clinically relevant set of findings, as well as its detailed voxellevel annotations that are essential for our explainability evaluation.

After an initial review of the thoracic findings available in ReXGroundingCT, a discussion with the supervising radiologist produced a shortlist of the following abnormalities that are clinically significant and have sufficient representation in the datasets:

1. Lung nodule
2. GroundGlass Opacity (GGO)

Both the volumes and their segmentation masks are formatted as NIfTI files. The masks are 3D volumetric segmentations ($H \times W \times D$) that align with the corresponding 3D CT scan, with their shape generally aligned with the scan's dimensions, as [512, 512, Depth]. For scans with multiple findings, masks can be structured as [F, H, W, D], where F represents the number of findings. Within a mask file, pixels are labeled with integer values corresponding to different entities for that finding, while 0 represents background.

2.2 Data Preprocessing

The data preprocessing pipeline consists of four major steps:

1. Resampling to isotropic spacing
2. Intensity windowing
3. Normalization
4. Patch extraction and vectorization

Each volume is resampled to 1.5 mm isotropic using trilinear interpolation to standardize the spatial resolution across scans. If the current spacing is already within 1% of target, this step is skipped. Each mask is also resampled to exactly match the resampled volume's spatial shape using nearestneighbour interpolation to preserve integer label values

In the second step, a standard window Hounsfield unit (HU) of [900, 200] was used to separate the lung window from the CT volume. Then, a linear rescaling to [0, 1] as float32 was applied to the windowed volume, to normalize the intensity values.

Finally, the preprocessed volumes are divided into 3D patches of size $16 \times 16 \times 16$ with 4,096 features per patch (24 mm^3 at 1.5 mm spacing). Normal and abnormal scans are processed in separate phases to ensure that the Dictionary Learning model first learns a representation of normal anatomy before being exposed to abnormalities.

In phase 1, patches are extracted on a regular non-overlapping grid with stride of 32 voxels from each normal scan. Only patches with at least 90% lung mask coverage (MIN_LUNG_COVERAGE = 0.90) are retained. All patches are labelled with class index 0 (normal).

In phase 2, using the abnormal scans, we slide a 3D patch window over each abnormality's bounding box with a fixed stride of $ABNORMAL\_STRIDE = 2$ voxels. All patches within the bounding box foreground are extracted regardless of specific overlap threshold. Patches with more than 50% zero voxels (background) are discarded. Each patch is directly labeled with the category being processed. For scans with multiple overlapping abnormalities, patches are extracted and labeled separately per category.

Both phases share the same maximum patches/scan budget; class balance is therefore determined by the normal/abnormal scan counts in the dataset, not a hard ratio All patches are assembled into X (4096 x N, float64) and H (N,) int64, then saved as .npz files.

2.3 Implementation of the LCKSVD Approach

To implement the Dictionary Learning approach for our study, we required a concrete implementation of the LCKSVD algorithm. [4] Currently, many of the available standard implementations of KSVD and LCKSVD are developed using MATLAB, which is not ideal for integration with our Pythonbased data processing and evaluation pipeline. Although A few opensource Python implementations LCKSVD are available, they either lacked an importable library, or did not provide the flexibility required for our study.

Therefore, we developed a custom implementation of the LCKSVD algorithm in Python and published it on PyPI, the official thirdparty software repository for Python, under the name "reppi." The implementation follows a modular design, consisting of four separate components:

1. SRC Utility Functions shared across Sparse Coding Algorithms
2. OMP Implementation of BatchOMP and OMPCholesky approaches for Sparse Coding
3. KSVD Implementation of KSVD Dictionary Learning Algorithm
4. LCKSVD Implementation of LCKSVD1 and LCKSVD2 Dictionary Learning Algorithms

This design also allows for flexibility in integrating alternative sparse coding methods in the future. The source code is available at github.com/ckekula/reppi.

This modular architecture enables straightforward integration of the algorithm through library imports, while providing the flexibility required to experiment with different hyperparameters and configurations. The implementations of the SRC, OMP, and LCKSVD modules are provided in Appendix A, Appendix B, and Appendix C, respectively.

2.4 Model Training and Inference

Prior to training, each patch vector is L2normalised to unit norm, ensuring that the sparse coding step operates on texture direction rather than absolute intensity magnitude. Patches with a norm below 1×10^{10}, corresponding to uniform zero regions with no signal, are discarded entirely.

The model is trained using LCKSVD2, a dictionary learning algorithm that jointly optimises three objectives: reconstruction accuracy, label consistency, and linear classification. Training proceeds in two stages: 15-iteration KSVD warmstart that initialises the dictionary on reconstruction error alone, followed by 30 LCKSVD2 iterations that refine it under the combined objective. This twostage procedure avoids poor local minima that arise when the labelconsistency and classifier terms are applied before the dictionary has converged to a geometrically meaningful initialisation.

The key hyperparameters are a dictionary size of K = 128 atoms, a sparsity constraint of T = 10 nonzero coefficients per patch (approximately 8% of K), a label-consistency weight of $\alpha = 4.0$, and a classifier weight of $\beta = 2.0$. On smaller datasets where the patch count N is insufficient to support 128 atoms, the dictionary size is adaptively capped at $max(8, N/2)$ and the sparsity constraint is scaled proportionally.

In practice, discriminative atoms are identified by selecting those atoms d_k whose classifier weight $|W[c, k]|$ exceeds the 75th percentile of all weights for class c. This filtering removes reconstruction-focused atoms that lack class-specific discriminative power, focusing the backprojection on class-relevant features.

At inference time, a new CT volume is loaded and preprocessed through the identical pipeline used during training. However, unlike training, instead of randomly sampling patches as in training, inference requires spatial exhaustiveness so that every region of the volume receives a classification score.

This is achieved by extracting patches on a dense regular grid with a stride of 8 voxels along each spatial axis (stride = PATCH_SIZE / 2 = 16 / 2 = 8). For a volume of size 512^3, this produces approximately $64^3 = 262144$ patches, each a 16^3 voxel window centred on a grid point. The overlap between adjacent patches (stride=8, patch_size=16) provides spatial coverage with 50% overlap, providing multiple independent score estimates per spatial location for the subsequent backprojection step.

2.5 BackProjecting ClassDiscriminative Atoms

The unique intuition we propose in our approach is to generate spatial contribution maps through backprojection. Because each atom in the dictionary D is associated with a specific class label through the matrix Q [4], the model can theoretically "explain" its decision for a test patch by examining which atoms were used in its sparse representation.

To generate a spatial contribution map for a class c, the algorithm identifies all atoms d_k that have been assigned to that class. For a query voxel or patch with sparse coefficients x, the contribution C_c is calculated as the partial reconstruction:

$$C_c = \sum_{k \in \text{Class } c} d_k x_k$$

This process backprojects the classspecific features into the original voxel space, creating a map that highlights the regions contributing to the detection of a specific abnormality. for classdiscriminative atoms operating on raw voxelspace inputs, the backprojection is an exact, closedform spatial attribution.

2.6 Deriving Explainability through Segmentation Masks

To evaluate the clinical accuracy of LCKSVD, these contribution maps are converted into binary segmentation masks for comparison with ground truth annotations. This conversion can be achieved through various thresholding methods.

GradCAM's approximation has two sources of imprecision — the gradient itself is a local linear approximation of a nonlinear function, and the spatial upsampling from the coarse feature map to input resolution introduces further interpolation error. LCKSVD's contribution map has only one source of imprecision: the thresholding step you apply to binarize it into a mask. The map itself — before thresholding — is exact given the sparse code and the dictionary.

Unlike the passive GradCAM approach, atom backprojection is generative. The atoms are what the model actually used to reconstruct and classify the signal. There's no proxy involved. The localization evidence is produced by the same computation that produced the classification, not by a separate posthoc analysis run afterwards.

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